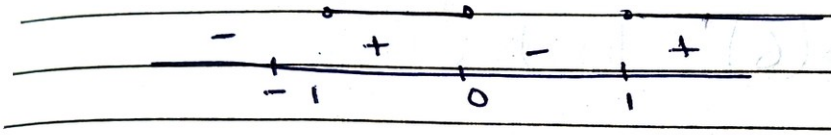
$$\frac{2x+2 - x^2 - x - 2}{x+1} < 0$$

$$\frac{2x - x^2 - x}{x+1} < 0$$

$$\frac{-x^2 + x}{x+1} < 0$$

$$\frac{x^2 - x}{x+1} > 0$$

$$\frac{x(x-1)}{x+1} > 0$$



→ Domain

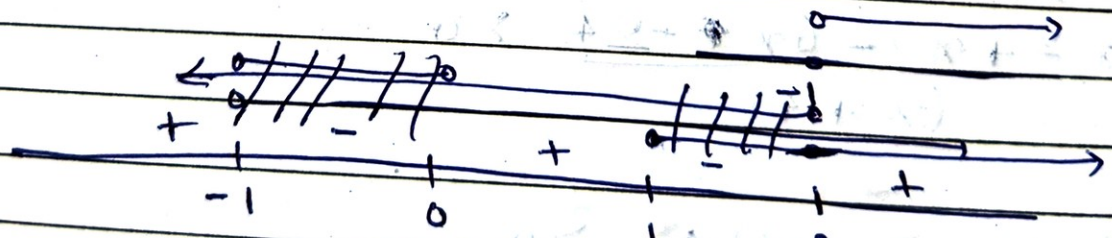
$$2 - x\left\{\log_{\frac{1}{4}}(2-x)\right\} > 0$$

$$-x > 0 + 2$$

$$(x < 2)$$

$$\frac{2}{x+1}\left[\log_{\frac{1}{4}}\frac{2}{x+1}\right] > 0$$

$$\frac{2}{x+1} > 0$$



$$x \in (-1, 0) \cup (1, 2)$$

Ex) $\log_{0.5}\left[\log_6 \frac{x^2+x}{x+4}\right] < 0$

$\Rightarrow \log_{0.5}\left[\log_6 \frac{x^2+x}{x+4}\right] < 0$

$\rightarrow \log_6\left[\frac{x^2+x}{x+4}\right] > (0.5)^0$

$\rightarrow \log_6\left[\frac{x^2+x}{x+4}\right] > 1$

$\rightarrow \frac{x^2+x}{x+4} > (6)^1$

$\rightarrow \frac{x^2+x}{x+4} - 6 > 0$

$\rightarrow \frac{(x^2+x) - 6(x+4)}{(x+4)} > 0$

$\rightarrow \frac{x^2+x-6x-24}{(x+4)} > 0$

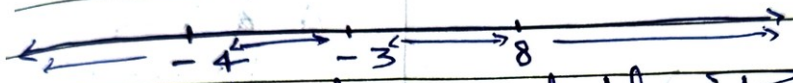
$\rightarrow \frac{x^2-5x-24}{x+4} > 0$

$\rightarrow \frac{x^2-8x+3x-24}{x+4} > 0$

$$\frac{x(x-8)+3(x-8)}{x+4} > 0$$

$$\frac{(x-8)(x+3)}{(x+4)} > 0$$

$x \Rightarrow +8, -3, -4$



4 intervals and we need the 'Positive intervals only' cause of the inequality $\Rightarrow (-\infty, -4), (-4, -3), (-3, 8), (8, \infty)$

$(+)^{ve}$ intervals $\rightarrow (-4, -3) \cup (8, \infty)$

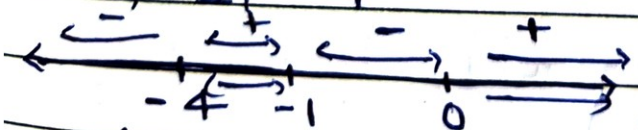
This intervals is for the log functions.

Now for the "$\left[\frac{x^2+x}{x+4}\right]$" which should be always greater than zero.

$$\frac{x^2+x}{x+4} > 0$$

$$\frac{x(x+1)}{(x+4)} > 0$$

$x \Rightarrow 0, -1, -4$



$(-4, -1) \cup (0, \infty)$

Intersection of $(-4, -1) \cup (0, \infty) \cap (-4, -3) \cup (8, \infty)$

$\Rightarrow (-4, -3) \cup (8, \infty)$

Ex) Solution Set of the inequality $2 - \log_2(x^2+3x) \geq 0$ is.

$\rightarrow 2 - \log_2(x^2+3x) \geq 0$

$\rightarrow \log_2(x^2+3x) \leq 2$

$\rightarrow \log_2(x^2+3x) \leq 2$

$\Rightarrow (x^2+3x) \leq (2)^2$

$\rightarrow x^2+3x \leq 4$

$\rightarrow x^2+3x-4 \leq 0$

$\rightarrow x^2+4x-x-4 \leq 0$

$\rightarrow x(x+4)-1(x+4) \leq 0$

$\rightarrow (x+4)(x-1) \leq 0$

$x \Rightarrow -4, 1$



$(-4, 1), (-\infty, -4)$

$(1, \infty) \Rightarrow (-4, 1)$

$x \Rightarrow -2$

$(-2)^2 + 3(-2) - 4$

$4 - 6 - 4$

$10 - 4 \Rightarrow +6$

$(-5)^2 + 3(-5) - 4$

$25 + 15 - 4$

$(-3)^2 + 3(-3) - 4$

$25 - 15 \Rightarrow 10 - 4 \Rightarrow +$

$\Rightarrow x^2 + 3x > 0$

$\Rightarrow x(x+3) > 0$

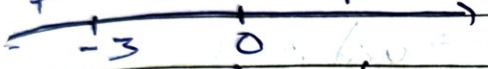
$\Rightarrow x = 0, -3$

$(-4)^2 + 3(-4)$

$16 - 12 \Rightarrow 4$

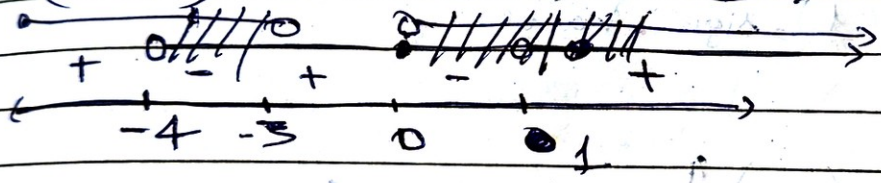
$(-2)^2 + 3(-2)$

$4 - 6 \Rightarrow -2$



$(-\infty, -3) \cup (0, \infty)$

$\Rightarrow (-4, 1) \cap (-\infty, -3) \cup (0, \infty)$



$\rightarrow [-4, -3)$

$\rightarrow [-4, -3)$ and $(0, 1]$

Ex) If $\log_{1/3}\left[\frac{3x-1}{x+2}\right]$ is less than 1 then "x" must lie in the interval.

$\rightarrow \log_{1/3}\left[\frac{3x-1}{x+2}\right] > 1$ → unity

less than (1), sign change

$\rightarrow \frac{3x-1}{x+2} < \left(\frac{1}{3}\right)^1$

$\rightarrow \frac{3x-1}{x+2} < \frac{1}{3}$

$\rightarrow \frac{3x-1}{x+2} - \frac{1}{3} < 0$

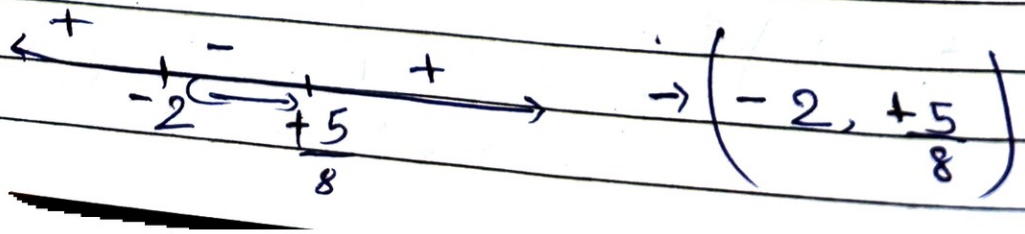
$\rightarrow \frac{3(3x-1) - (x+2)}{(x+2)3} < 0$

$\rightarrow \frac{9x-3-x-2}{3x+6} < 0$

$\rightarrow \frac{8x-5}{3x+6} < 0$

$\rightarrow x \Rightarrow \frac{+5}{8}$ $\quad -0.625$

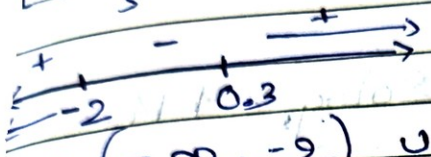
$x \Rightarrow \frac{-6}{8} \Rightarrow -2$



$\rightarrow \left(-2, \frac{+5}{8}\right)$
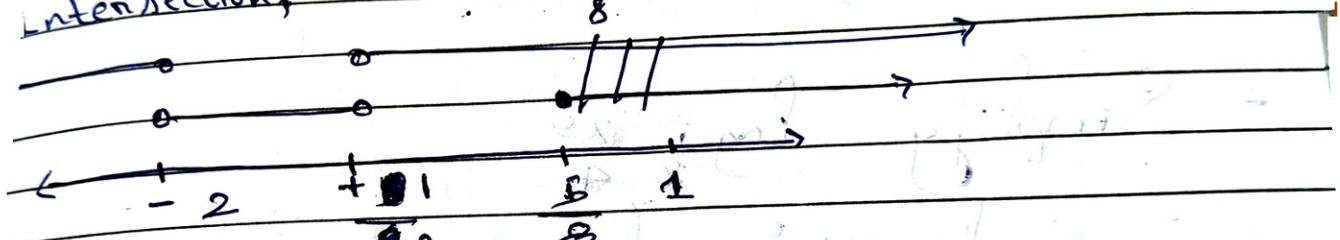
$\frac{3x-1}{x+2} \geq 0$

$\rightarrow \frac{1}{3}, -2$



$\rightarrow (-\infty, -2) \cup (\frac{1}{3}, \infty)$

Intersection; $\rightarrow (-2, \frac{-5}{8}) \cap (-\infty, -2) \cup (\frac{1}{3}, \infty)$



$\rightarrow (-\infty, -2) \cup (\frac{1}{3}, \frac{5}{8})$

$\rightarrow (-\infty, -2) \cup (\frac{1}{3}, \frac{5}{8})$

$\rightarrow (-\infty, -2) \cup (+\frac{5}{8}, \infty)$

Ex) $(\log_2 x)^4 - \left[\log_{1/2} \frac{x^5}{4}\right]^2 - 20\log_2 x + 148 < 0$

$\Rightarrow (\log_2 x)^4 - \left[\log_{1/2} \frac{x^5}{4}\right]^2 - 20\log_2 x + 148 < 0$

$\rightarrow$ Let $\log_2 x = t$

$\rightarrow$ Simplify $\log_{\frac{1}{2}} \frac{x^5}{4}$

$\Rightarrow \log_{\frac{1}{2}} x^5 - \log_{\frac{1}{2}} 4$

$\Rightarrow 5\log_{\frac{1}{2}} x - (-2)$

$\Rightarrow -5\log_2 x + 2$

$\rightarrow (\underbrace{\log_2 x}_{t})^4 - \left[-5\underbrace{\log_2 x}_{t} + 2\right]^2 - 20\underbrace{\log_2 x}_{t} + 148 < 0$

$\rightarrow (t)^4 - [(-t) + 2]^2 - 20t + 148 < 0$

$\rightarrow t^4 - [(-t)^2 + 2(2)(-t) + (2)^2] - 20t + 148 < 0$

$\rightarrow t^4 - [25t^2 - 20t + 4] - 20t + 148 < 0$

$\rightarrow t^4 - 25t^2 + 20t - 4 - 20t + 148 < 0$

$\rightarrow t^4 - 25t^2 + 144 < 0$

let $\log_2 x = t^2 \Rightarrow y$

$\Rightarrow (t^2)^2 - 25(t^2) + 144 < 0$

$\Rightarrow y^2 - 25y + 144 < 0$

$\Rightarrow y^2 - 9y - 16y + 144 < 0$

$\Rightarrow y(y-9) - 16(y-9) < 0$

$\Rightarrow (y-9)(y-16) < 0$

$\Rightarrow y \Rightarrow 9, 16$

Now

$\Rightarrow 9 < y^2 < 16$

$\Rightarrow 9 < t^2 < 16$

$\Rightarrow t^2 > 9$

$\Rightarrow t > 3$ (or) $t < -3$ (and) $t > -4$ (or) $t < 4$

$\Rightarrow \log_2 x > 3$ (or) $\log_2 x < -3$ and $\log_2 x > -4$ (or) $\log_2 x < 4$

1) $\log_2 x > 3$

$x > (2)^3$

$x > 8$

2) $\log_2 x < -3$

$x < (2)^{-3}$

$x < \frac{1}{8}$

(3) $\log_2 x > -4$

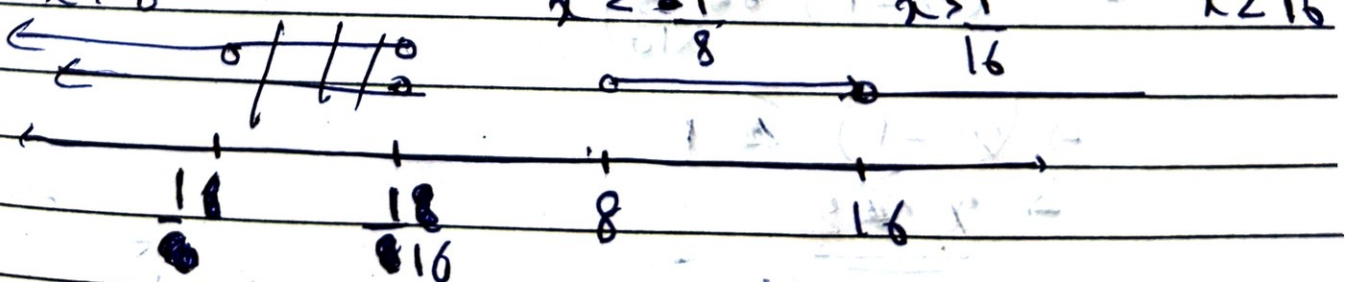
$x > (2)^{-4}$

$x > \frac{1}{16}$

4) $\log_2 x < 4$

$x < (2)^4$

$x < 16$



$\Rightarrow \left(\frac{1}{16}, \frac{1}{8}\right) \cup (8, 16)$

Home - Work

Ex) If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval

$\rightarrow \log_{0.3}(x-1) < \log_{0.09}(x-1)$

$\rightarrow \log_{0.3}(x-1) - \log_{0.09}(x-1) < 0$

$\rightarrow \log_{0.3}(x-1) - \log_{\frac{9}{100}}(x-1) < 0$

$\rightarrow \log_{\frac{3}{10}}(x-1) - \log_{(\frac{3}{10})^2}(x-1) < 0$

$\rightarrow \log_{\frac{3}{10}}(x-1) - \frac{1}{2}\log_{0.3}(x-1) < 0$

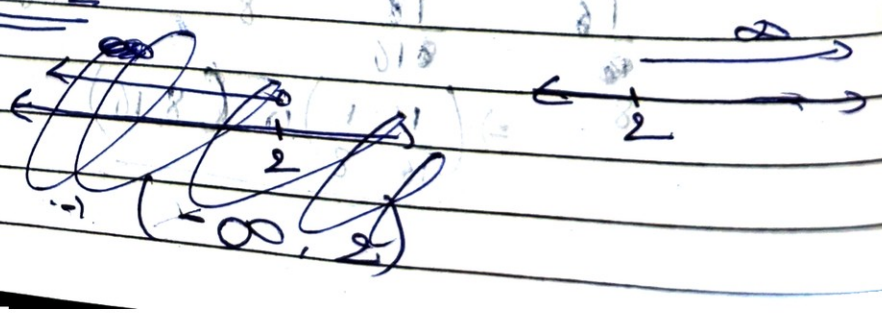
$\rightarrow 2\log_{0.3}(x-1) - \log_{0.3}(x-1) < 0$

$\rightarrow \log_{0.3}(x-1) < 0$

$\rightarrow (x-1) > (\frac{3}{10})^0$

$\rightarrow (x-1) > 1$

$\rightarrow x > 2$



→ $x - 1 \geq 0$

→ $x \geq 1$



→ $(1, 2)$, $(2, \infty)$ final ans

x) The Solution of the inequality $\log_{\frac{5}{8}}\left[2x^2 - x - \frac{3}{8}\right] \geq 1$ is

→ $\log_{\frac{5}{8}}\left[2x^2 - x - \frac{3}{8}\right] \geq 1$

→ $\left[2x^2 - x - \frac{3}{8}\right] \leq \frac{5}{8}$

→ $16x^2 - 8x - 3 \leq 5$ → forgot to change the sign of inequality

→ $16x^2 - 8x - 3 + 3 \leq 5$

→ $16x^2 - 8x - 8 \leq 0$

→ $8(2x^2 - x - 1) \leq 0$

→ $2x^2 - x - 1 \leq 0$ → changed it

→ $2x^2 - x - 1 \geq 0$

→ $2x^2 - 2x + x - 1 \geq 0$

→ $2x(x - 1) + 1(x - 1) \geq 0$

→ $(x - 1)(2x + 1) \geq 0$

→ $x = 1$ and $-\frac{1}{2}$



→ $\left[-\frac{1}{2}, 1\right]$

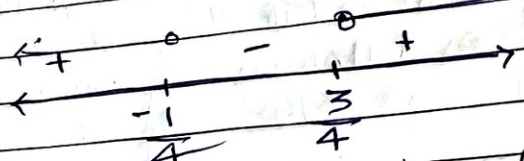
$\Rightarrow$

$2x^2 - x - \frac{3}{8} > 0$

$16x^2 - 8x - 3 > 0$

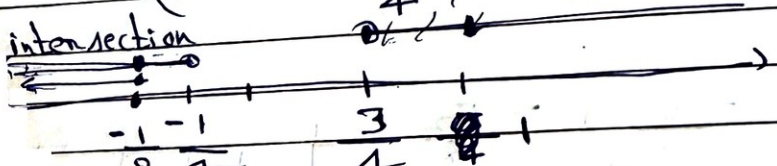
$[4x+1][4x-3] > 0$

$\rightarrow x \Rightarrow \frac{-1}{4}$ and $\frac{3}{4}$



$\Rightarrow \left(-\infty, \frac{-1}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$

$\Rightarrow$ intersection



$\rightarrow \left[\frac{-1}{2}, \frac{-1}{4}\right) \cup \left(\frac{3}{4}, 1\right]$

Logarithmic Inequalities (With Variable base)

∴ $f(x) \Rightarrow \log_a x$

1) $x > 0$
2) $a > 0$
3) $a \neq 1$

⇒ base > 1 ⇒ same
⇒ base < 1 ⇒ change

∴ $\log_x \left(2x - \frac{3}{4}\right) > 2$

⇒ Two cases Whenever the base is Variable

Case-1 ⇒ $0 < x < 1$ (or) Case-2 $x > 1$

→ $\log_x \left(2x - \frac{3}{4}\right) > 2$

→ $2x - \frac{3}{4} > x^2$

→ $2x - \frac{3}{4} < x^2$

→ $8x - 3 < 4x^2$

→ $0 < 4x^2 - 8x + 3$

→ $4x^2 - 8x + 3 > 0$

→ $4x^2 - 6x - 2x + 3 > 0$

→ $4x^2 - 2x - 6x + 3 > 0$

→ $2x(2x-1) - 3(2x-1) > 0$

→ $(2x-1)(2x-3) > 0 \Rightarrow$

Case-2 $x > 1$

→ $2x - \frac{3}{4} > x^2$

→ $4x^2 < 8x - 3$

→ $4x^2 - 8x - 3 < 0$

→ $(2x-1)(2x-3) < 0$

Case -1

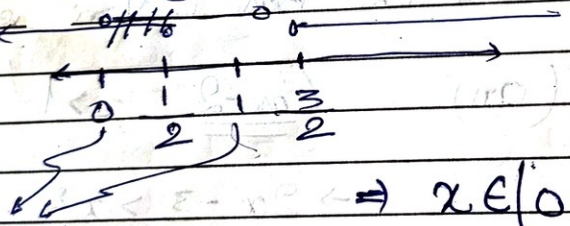
Note:-

We took a Case -1 where the "x" is in b
$0 < x < 1$ and after solving the expression
will get an answer of "x" from the function
that we will take intersection of the Case th
we assumed and the answer of the function an
we will form intervals.

$\rightarrow 0 < x < 1$

$(2x-3)$ and $(2x-1) > 0$

$x \Rightarrow \frac{3}{2}$ and $\frac{1}{2}$



from the case $\Rightarrow x \in \left(0, \frac{1}{2}\right)$

Case -2

$\rightarrow$ Same as we did in Case-1

$\Rightarrow x > 1$

$(x-3)(2x-1) < 0$



$\Rightarrow x \in \left(\frac{1}{2}, \frac{3}{2}\right)$

Note:
Then after getting answers for both Cases We Will Do the union of both of the Case.

$x \in \left(0, \frac{1}{2}\right)$, $x \in \left(1, \frac{3}{2}\right)$

⇒ final answer :-

$$\left[\text{Case}(1) \cup \text{Case}(2)\right] \cap \text{Domain}$$

⇒ Domain,

$$\frac{2x - 3}{4} > 0$$

$$8x - 3 > 0$$

$$8x > 3$$

$$x > \frac{3}{8}$$

off off



$\Rightarrow \left(0, \frac{1}{2}\right) \cup \left(1, \frac{3}{2}\right)$

$\rightarrow \left(\frac{3}{8}, \frac{1}{2}\right) \left(1, \frac{3}{2}\right)$

Ex) $\log_x(x^3 - x^2 - 2x) < 3$

→ Case - 1

$0 < x < 1$

$\log_x(x^3 - x^2 - 2x) < 3$

$(x^3 - x^2 - 2x) < x^3$

$(x^3 - x^2 - 2x - x^3) > 0$

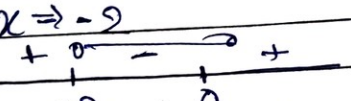
$-x^2 - 2x > 0$

$x^2 + 2x < 0$

$x(x+2) < 0$

$x \Rightarrow 0$

$x \Rightarrow -2$



$\Rightarrow (-2, 0)$ {No Solution Case}

$0 < x < 1$ ~~#-#~~

$x \in (0, 1)$ Case

⇒ Case - 2

→ $x > 1$

→ $x^3 - x^2 - 2x < x^3$

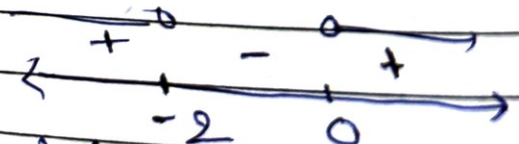
→ $x^3 - x^2 - 2x - x^3 < 0$

→ $-x^2 - 2x < 0$

→ $x^2 + 2x > 0$

→ $x(x+2) > 0$

→ $x \Rightarrow 0, -2$



$\Rightarrow (-\infty$

I

∴ Case 1 and Case 2 ⇒ Union

⇒ $(-2,0) \cup (-\infty,-2) \cup (0,\infty)$

Domain,

⇒ $x^3 - x^2 - 2x > 0$

→ $x^3 - x^2 - 2x > 0$

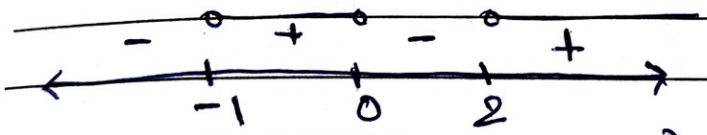
→ $x(x^2 - x - 2) > 0$

→ $x(x^2 - 2x + x - 2) > 0$

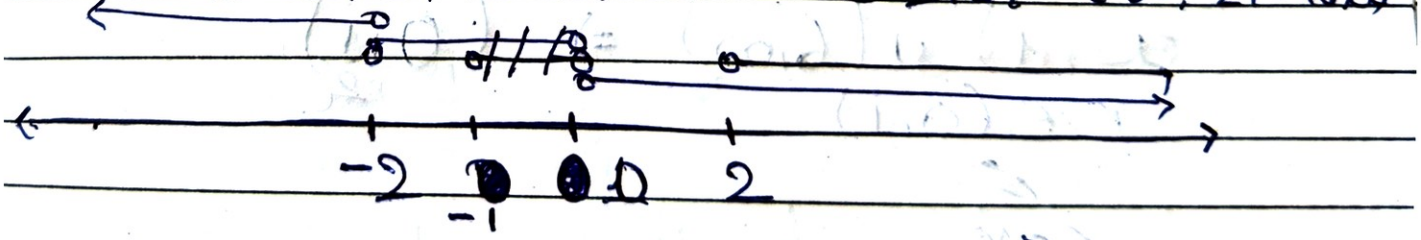
→ $x(x(x-2) + 1(x-2)) > 0$

→ $((x-2)(x+1)x > 0$

$x \Rightarrow 0, 2, -1$



⇒ $(-1,0) \cup (2,\infty) \cap (-2,0) \cup (-\infty,-2) \cup (0,\infty)$



→ ~~$(-1,0) \cup (0,\infty)$~~ → ~~$(-1,0)$~~ ~~$\cap$~~ $(2,\infty)$

~~$(-1,2)$~~

Ex) $\log_{2x}(x^2-5x+6) < 1$

→ Case - 1

$\log_{2x}(x^2-5x+6) < 1$

$0 < x < 1$, $0 < 2x < 1$, $0 < x < \frac{1}{2}$

$\log_{2x}(x^2-5x+6) < 1$

$(x^2-5x+6) < (2x)^1$

$x^2-5x+6-2x > 0$

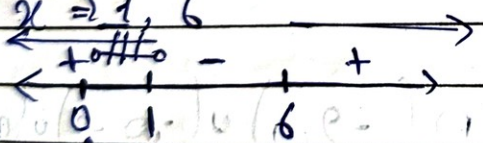
$x^2-7x+6 > 0$

$x^2-6x-x+6 > 0$

$x(x-6)-1(x-6) > 0$

$(x-1)(x-6) > 0$

$x = 1, 6$



$(-\infty, 1) \cup (6, \infty) \Rightarrow (0, \frac{1}{2})$

$x \in (0, 1)$

∈

case

Case - 2

$\log_{2x}(x^2 - 5x + 6) < 1$

$\Rightarrow 2x > 1$

$\Rightarrow x^2 - 5x + 6 < 2x$

$\Rightarrow x^2 - 5x + 6 - 2x < 0$

$\Rightarrow (x-1)(x-6)$



$\Rightarrow (1, 6)$

$\Rightarrow (1,6) \cup (0, \frac{1}{2})$

Case 1 and 2

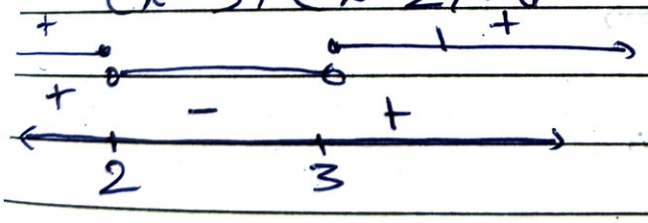
Domain :-

$x^2 - 5x + 6 \geq 0$

$x^2 - 3x - 2x + 6 \geq 0$

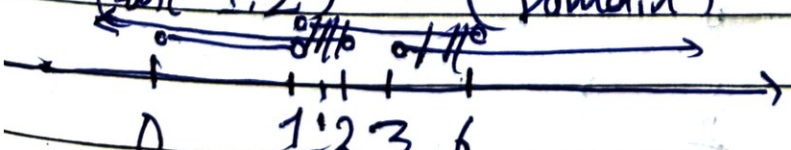
$x(x-3) - 2(x-3) \geq 0$

$(x-3)(x-2) \geq 0$



$\Rightarrow (-\infty, 2) \cup (3, \infty)$

$\Rightarrow$ (Case 1,2) $\cap$ (Domain)



$\rightarrow (0, \frac{1}{2}) \cup (1, 2)$

Ex) $\log_{x^2}(2+x) < 1$

→ Case - 1

→ $0 < x^2 < 1$

→ $0 < x^2 - 1 < 0$

→ $(x+1)(x-1) < 0$

→ $x \in (-1, 0) \cup (0, 1)$

→ $\log_{x^2}(2+x) < 1$

$(2+x) < (x^2)^1$

$2 + x < x^2$

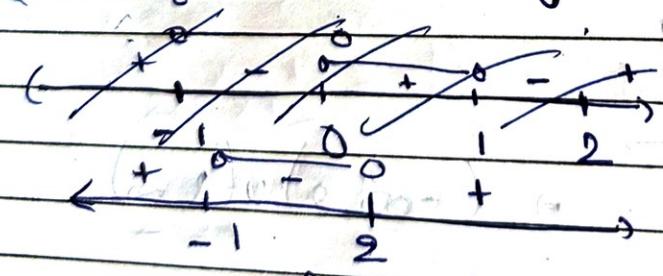
$x^2 - x - 2 > 0$

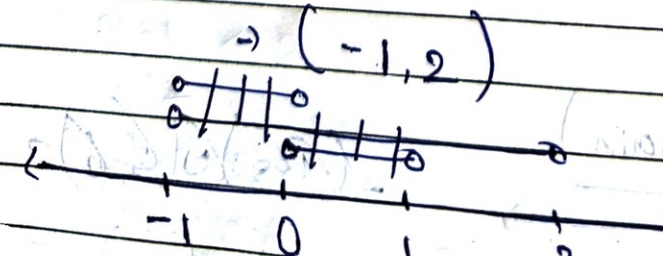
$x^2 - x - 2 < 0$

$x^2 - 2x + x - 2 < 0$

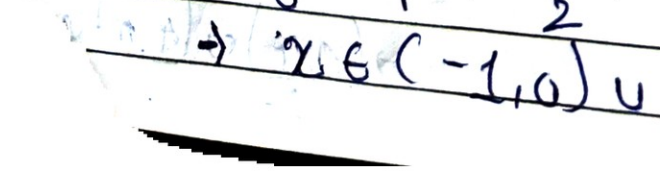
$x(x-2) + 1(x-2) < 0$

$(x-2)(x+1) < 0$





→ $(-1, 2)$



→ $x \in (-1, 0) \cup (0, 1)$